

Формирование временной корневой файловой системы

Задание 1

Создание файла `init.c` с помощью `nano`.

```
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>

int main(void)
{
    printf("\nCOMPLETE!\n");
    sleep(10);
}
```

Компиляция `init` для ARM с помощью кросс компилятора:

```
pavel@pavel-MCLF-XX:~/Рабочий стол/kernel$ arm-linux-gnueabi-gcc -static init.c -o init
pavel@pavel-MCLF-XX:~/Рабочий стол/kernel$
```

Упаковка в архив:

```
pavel@pavel-MCLF-XX:~/Рабочий стол/kernel$ echo init | cpio -o -H newc | gzip > initramfs.cpio.gz
829 блоков
pavel@pavel-MCLF-XX:~/Рабочий стол/kernel$
```

Запуск ядра и работоспособность 10сек:

```

pavel@pavel-MCLF-XX:~/Рабочий стол/kernel$ qemu-system-arm -M vexpress-a9 -m 512M -kernel z
Image -dtb vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append "console=ttyAMA0 root=/de
v/ram rdinit=/init" -nographic
Booting Linux on physical CPU 0x0
Linux version 6.14.0 (pavel@pavel-MCLF-XX) (arm-linux-gnueabi-hf-gcc (Ubuntu 13.3.0-6ubuntu2
24.04-1) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #3 SMP Tue Jun 10 16:10:07 2025

Run /init as init process

COMPLETE!
input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:mothe
rboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/se
rio1/input/input1
drm-clcd-pl111 10020000.clcd: DVI muxed to daughterboard 1 (core tile) CLCD
drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111
input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000:motherboard-bus
@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10006000.kmi/serio0/input
/input2
drm-clcd-pl111 10020000.clcd: DVI muxed to daughterboard 1 (core tile) CLCD
drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111

Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
CPU: 0 UID: 0 PID: 1 Comm: init Not tainted 6.14.0 #3
Hardware name: ARM-Versatile Express
Call trace:
 unwind_backtrace from show_stack+0x10/0x14
 show_stack from dump_stack_lvl+0x54/0x68
 dump_stack_lvl from panic+0x10c/0x350
 panic from make_task_dead+0x0/0x16c
---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000 ]---

```

Задание 2

Скачали BusyBox с официального сайта и распаковали

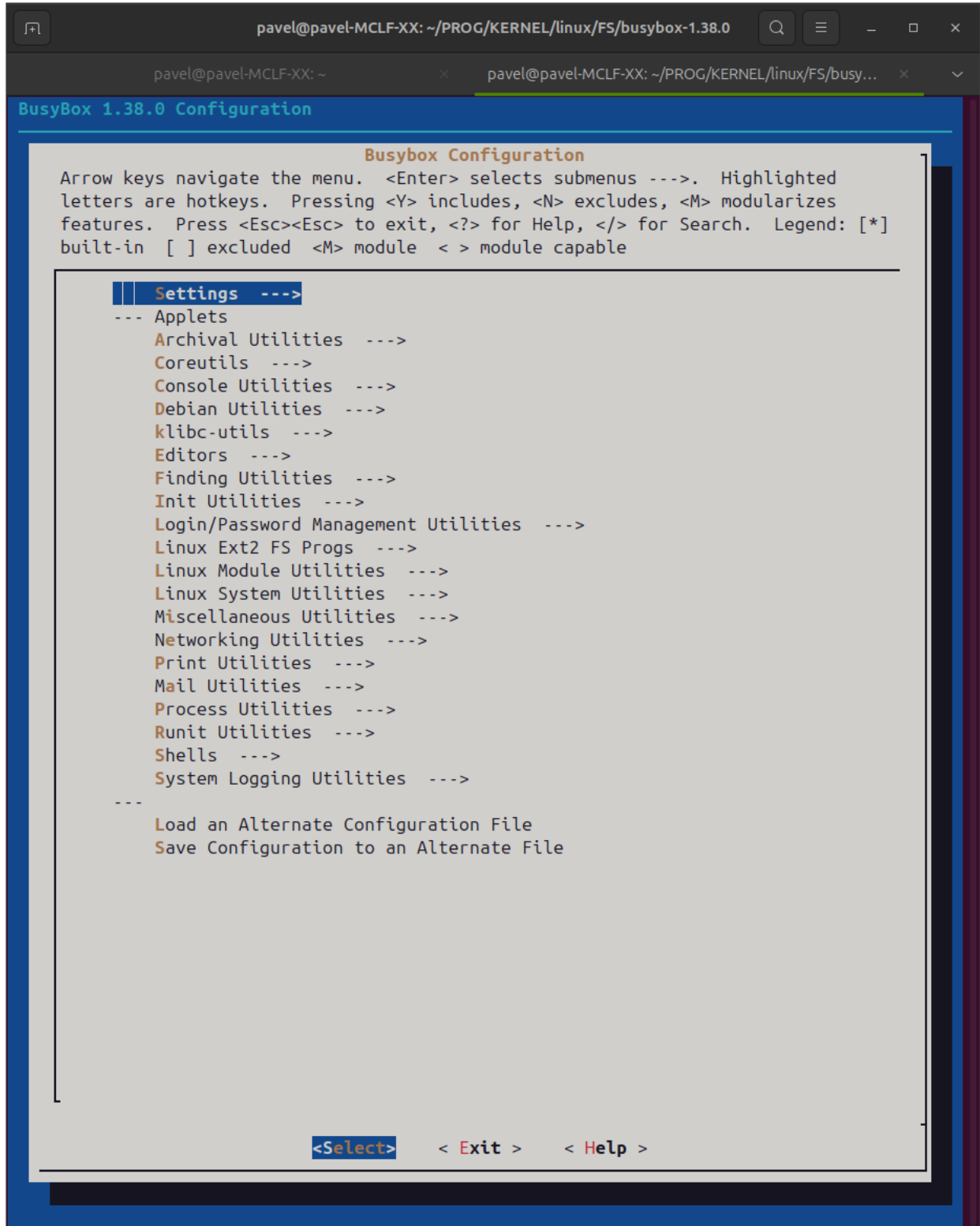
Сформировали конфиг:

```

pavel@pavel-MCLF-XX:~$ cd /home/pavel/PROG/KERNEL/linux/FS/busybox-1.38.0
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux/FS/busybox-1.38.0$ ARCH=arm make defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/basic/split-include
HOSTCC scripts/basic/docproc
GEN include/applets.h
GEN include/usage.h
GEN loginutils/Kbuild
GEN loginutils/Config.in
GEN findutils/Kbuild

```

Редакция конфига



The image shows a terminal window with the title bar "pavel@pavel-MCLF-XX: ~/PROG/KERNEL/linux/FS/busybox-1.38.0". The terminal displays the "BusyBox 1.38.0 Configuration" menu. At the top, a blue header bar contains the text "BusyBox 1.38.0 Configuration". Below this, the menu is titled "Busybox Configuration" in orange. A paragraph of instructions follows: "Arrow keys navigate the menu. <Enter> selects submenus --->. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [] excluded <M> module < > module capable". The main menu is a list of submenus, each preceded by "--->". The "Settings" submenu is highlighted with a blue background and white text. At the bottom of the menu, there are three options: "<Select>", "< Exit >", and "< Help >".

```
pavel@pavel-MCLF-XX: ~/PROG/KERNEL/linux/FS/busybox-1.38.0
BusyBox 1.38.0 Configuration

Busybox Configuration
Arrow keys navigate the menu. <Enter> selects submenus --->. Highlighted
letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes
features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*]
built-in [ ] excluded <M> module < > module capable

---> Settings --->
--- Applets
    Archival Utilities --->
    Coreutils --->
    Console Utilities --->
    Debian Utilities --->
    klibc-utils --->
    Editors --->
    Finding Utilities --->
    Init Utilities --->
    Login/Password Management Utilities --->
    Linux Ext2 FS Progs --->
    Linux Module Utilities --->
    Linux System Utilities --->
    Miscellaneous Utilities --->
    Networking Utilities --->
    Print Utilities --->
    Mail Utilities --->
    Process Utilities --->
    Runit Utilities --->
    Shells --->
    System Logging Utilities --->
---
    Load an Alternate Configuration File
    Save Configuration to an Alternate File

<Select> < Exit > < Help >
```

```
[*] Build static binary (no shared libs)
```

```
[ ] Force NOMMU build
```

```
(arm-linux-gnueabihf-) Cross compiler prefix
```

Networking Utilities

Now keys navigate the menu. <Enter> selects submenus --->. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [] excluded <M> module < > module capable

```
↑(-)
```

```
[*] Use internal resolver code instead of libc
```

```
[*] Enable long options
```

```
[*] ntpd (23 kb)
```

```
[*] Make ntpd usable as a NTP server
```

```
[*] Make ntpd understand /etc/ntp.conf
```

```
[*] Support md5/sha1 message authentication codes
```

```
[*] ping (10 kb)
```

```
[*] ping6 (11 kb)
```

```
[*] Enable fancy ping output
```

```
[*] pscan (6.2 kb)
```

```
[*] route (9 kb)
```

```
[*] slattach (6.3 kb)
```

```
[*] ssl_client (28 kb)
```

```
[*] ssl_server (test TLS server)
```

```
[ ] tc (8.3 kb)
```

```
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux/FS/busybox-1.38.0$ make -j12
```

```
SPLIT include/autoconf.h -> include/config/*
```

```
GEN include/bbconfigopts.h
```

```
GEN include/common_bufsiz.h
```

```
GEN include/embedded_scripts.h
```

```
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux/FS/busybox-1.38.0$ make install  
/install //bin/ash -> busybox
```

После этого следуя этапам из лекции у меня ядро не поднималось и я решил взять информацию из интернета:

Смонтировал devtmpfs - специальная файловая система ядра, которая сама автоматически создает все нужные файлы устройств (tty, null, console и т.д.) прямо при старте

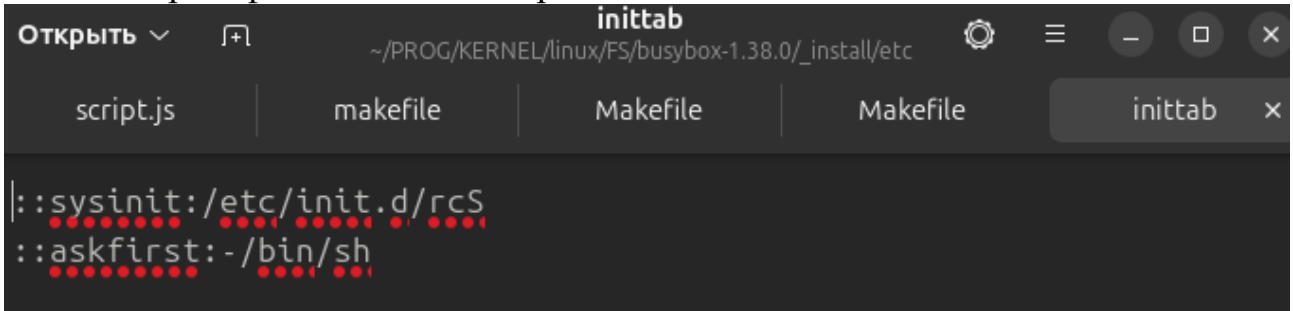
в _install создал etc/init.d/rcS

в rcS :

```
#!/bin/sh
mount -t devtmpfs none /dev
mount -t proc none /proc
mount -t sysfs none /sys
echo "Система на базе собственного BusyBox ARM успешно загружена!"
```

chmod +x etc/init.d/rcS

так же открыл файл etc/inittab и прописал



```
inittab
~ /PROG/KERNEL/linux/FS/busybox-1.38.0/_install/etc

script.js | makefile | Makefile | Makefile | inittab x

|:sysinit:/etc/init.d/rcS
::askfirst:/bin/sh
```

Создаем архивы и копируем его в нашу папку kernel, перед этим удалим архив 1 задания.

```
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux/FS/busybox-1.38.0/_install$ find . | cpio -o -H newc | gzip >
initramfs.cpio.gz
3159 блоков
pavel@pavel-MCLF-XX:~/PROG/KERNEL/linux/FS/busybox-1.38.0/_install$
```

Запуск:

```
pavel@pavel-MCLF-XX:~/Рабочий стол/kernel$ qemu-system-arm -M vexpress-a9 -m 512M -kernel zImage -dtb
vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append "console=ttyAMA0 root=/dev/ram rdinit=/init" -n
ographic
```

Работоспособность:

Система на базе собственного BusyBox ARM успешно загружена!

Please press Enter to activate this console. input: ImExPS/2 Generic Explorer Mouse as /dev/input/mouse0
rm/bus@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:i
0000000/10007000.kmi/serio1/input/input2

drm-clcd-pl111 10020000.clcd: DVI muxed to daughterboard 1 (core tile) CLCD

drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111

drm-clcd-pl111 10020000.clcd: DVI muxed to daughterboard 1 (core tile) CLCD

drm-clcd-pl111 10020000.clcd: initializing Versatile Express PL111

i2c 0-0039: deferred probe pending: sii902x: Failed to find remote bridge

amba 1000f000.wdt: deferred probe pending: (reason unknown)

amba 10020000.clcd: deferred probe pending: (reason unknown)

amba 100e0000.memory-controller: deferred probe pending: (reason unknown)

amba 100e1000.memory-controller: deferred probe pending: (reason unknown)

amba 100e5000.watchdog: deferred probe pending: (reason unknown)

~ #

~ #

~ # ls

bin dev etc init root sbin usr

~ #